THE MILKY WAY

ALL THE STARS WE CAN SEE IN THE SKY LIE WITHIN THE CONFINES OF OUR HOME GALAXY, THE MILKY WAY. THIS VAST STAR SYSTEM, CONTAINING HUNDREDS OF BILLIONS OF STARS, IS A BARRED SPIRAL WITH A COMPLEX STRUCTURE AND IS ABOUT 120,000 LIGHT-YEARS ACROSS.

The Milky Way's visible stars form a disk centered on a bulging hub. Despite its vast diameter, the disk averages only a thousand light-years deep. From our point of view on Earth, we see many more stars looking across the plane of the disk than when we look "up" or "down" from the plane and out into intergalactic space. This is why we see our galaxy as a broad band whose countless faint and distant stars merge together in a milky band of light.

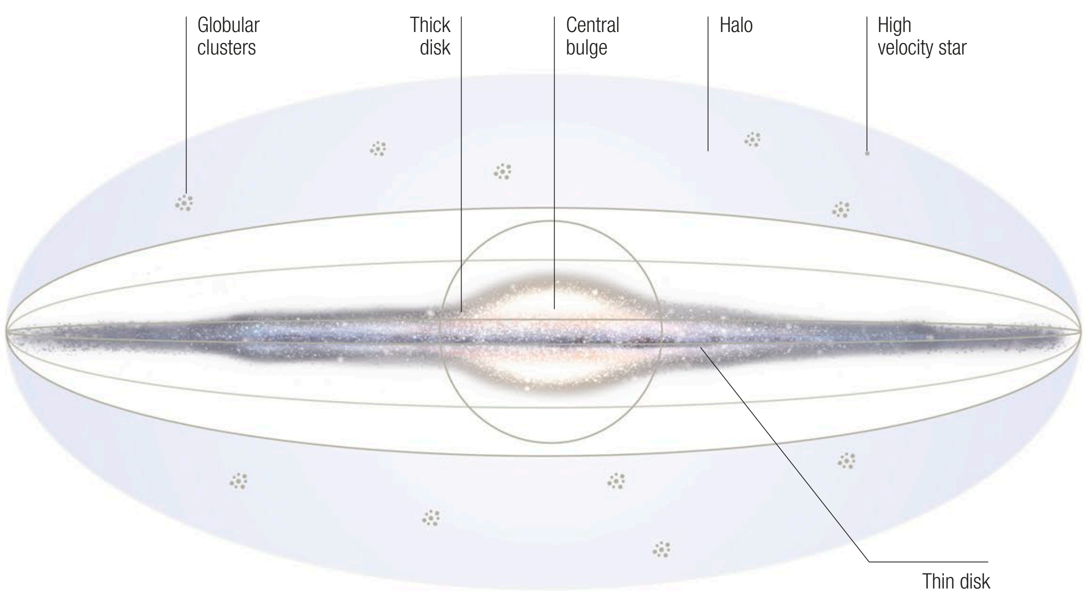
The central bulge of the Milky Way is dominated by low-mass, red and yellow stars with a high metallicity (see p.29), but the surrounding disk is filled with gas, dust, and younger stars. As with all spirals, stars are scattered across the disk, but the brightest are concentrated into the spiral arms. Stars orbit at different rates depending on their distance from the hub, so the arms cannot be permanent structures. Instead, they stand out because they are the active regions of star formation. Here, stars are born, and the most massive and luminous among them pass through their short life cycles before their orbits can carry them out into the wider disk.

Spiral arms

Astronomers have recently confirmed that the Milky Way is a barred spiral galaxy. Its central hub is crossed by a rectangular bar of stars some 27,000 light-years in length. Our galaxy's spiral arms are the result of stars, gas, and dust moving in and out of a spiral-shaped "traffic jam" called a density wave (see opposite).

The latest evidence suggests that the **Milky Way has four spiral arms**—two major and two minor—with distinct differences among their stars

Heart of the Milky Way
The central regions of our galaxy are hidden behind intervening star clouds and dust lanes, but X-rays and infrared can pierce the veil to reveal complex structures, giant star clusters, and an enormous black hole with the mass of several million Suns (embedded in the bright gas cloud to right of the image).



◁ **Cross section of the Milky Way**
Seen from the side, the Milky Way consists of a disk of stars around a bulging hub some 8,000 light-years across. A broad halo region above and below the galaxy appears largely empty, but is home to globular star clusters, as well as stray high-velocity stars and hot gas ejected from the galactic plane.